INTERNATIONAL MARITIME ORGANIZATION

E

SUB-COMMITTEE ON NAVIGATION, COMMUNICATIONS AND SEARCH AND RESCUE
13th session
Agenda item 3

NCSR 13/3/6
20 March 2026
Original: ENGLISH
Pre-session public release: ☒

**ROUTEING MEASURES AND SHIP REPORTING SYSTEMS**

**Amendments to the traffic separation scheme "In the approaches to the Cape Fear River"**

**Submitted by the United States**

**SUMMARY**

| | |
|---|---|
| *Executive summary:* | This document contains a proposal for the establishment of a new precautionary area at the offshore terminus of the traffic separation scheme *In the approaches to the Cape Fear River*. This proposed amendment should maintain traditional shipping routes and continue to ensure navigation lanes remain free of fixed structures. |
| *Strategic direction, if applicable:* | 7 |
| *Output:* | 7.22 |
| *Action to be taken:* | Paragraph 24 |
| *Related documents:* | *Ships' Routeing*, 2019 edition; SOLAS regulation V/10; resolution A.572(14), as amended; COLREG.2/Circ.55, COLREG.2/Circ.55/Corr.1; MSC.1/Circ.1060, as amended; NCSR 13/3/4, NCSR 13/3/5, NCSR 13/3/6 and NCSR 13/INF.8 |


## Introduction

1 The United States proposes to amend the traffic separation scheme (TSS) *In the approaches to Cape Fear River* adopted by IMO in 2004, as disseminated by COLREG.2/Circ.55 as corrected by COLREG.2/Circ.55/Corr.1. The existing TSS consists of a precautionary area, two traffic lanes, and a separation zone. Geographical coordinates and a chartlet depicting the TSS are provided in IMO *Ships' Routeing*, edition 2019.

## Summary

2 The United States proposes to establish a new precautionary area at the offshore terminus of the TSS *In the approaches to the Cape Fear River*. This proposed precautionary area would be an approximately 75-square mile area and would apply to all ships navigating through the area. Figure 1 provides a side-by-side comparison of the current TSS and with the proposed amendment.

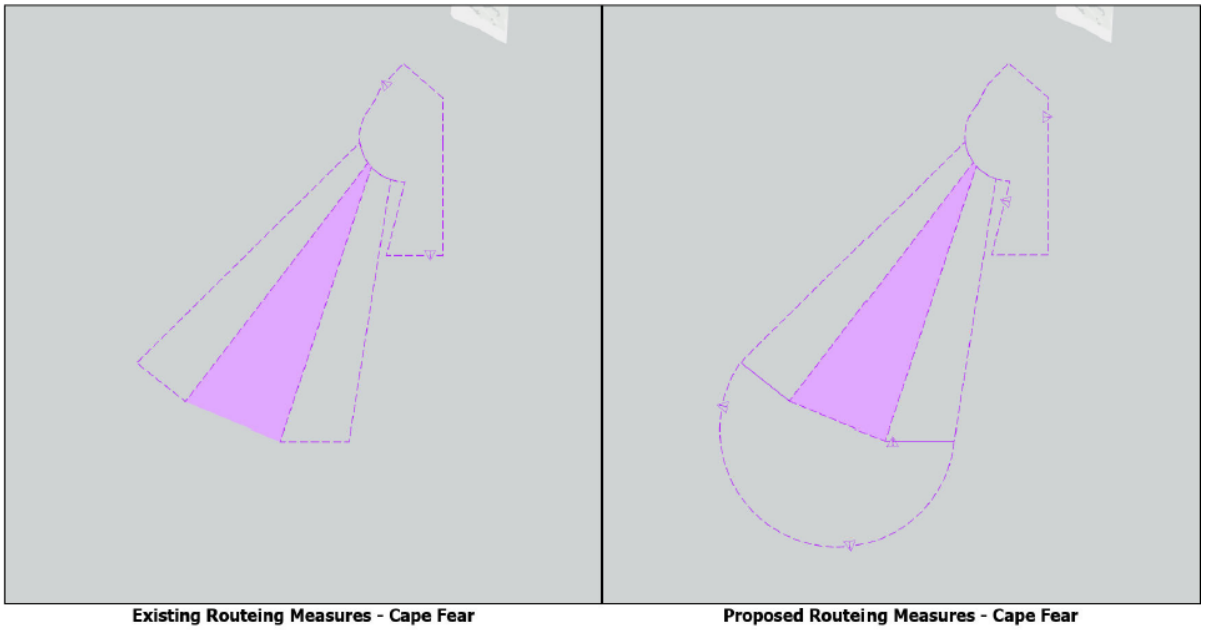



**Figure 1: Proposed amendments to the TSS in the approaches to the Cape Fear River**

3 The United States has placed particular attention on expanded oil, natural gas, hydrokinetic, solar, nuclear, aquaculture and other non-mineral energy resources on the outer continental shelf (OCS). In identifying viable locations in the United States exclusive economic zone (EEZ) for energy production, the United States has also identified safe access routes, or "fairways," for the movement of vessel traffic proceeding to or from the United States ports, in which no artificial island or fixed structure, whether temporary or permanent, will be permitted. As such, the designation of a fairway recognizes the paramount right of navigation over all other uses in the designated areas. The United States expects shipping transits along the Atlantic coast to consolidate into these designated fairways. Two of these routes will converge at the offshore terminus of the TSS *In the approaches to the Cape Fear River*. By establishing a new precautionary area at the offshore terminus of the TSS, this proposal addresses the meeting of vessel traffic at a critical intersection, requiring mariners to navigate with particular caution in the designated area. A precautionary area is the preferred routeing measure as it provides a charted identifiable area where ships must navigate with particular caution. A direction of traffic flow was considered, but determined to be unnecessary and inconsistent with all other precautionary areas throughout the United States. This proposal is preventative in nature and will help provide predictable navigation corridors to accommodate traditional shipping routes along with the evolving offshore energy infrastructure, thereby improving operational safety in increasingly congested areas along the United States Atlantic coast. This system will continue to apply to all ships transiting the area. All other existing schemes remain unchanged. Further discussion on traffic considerations is provided in paragraphs 8 and 9.

4 In addition, this document presents all geographical coordinates in IMO *Ships' Routeing* 2019 edition description of the TSS *In the approaches to the Cape Fear River* converted from the existing North American 1983 geodetic datum to the World Geodetic System of 1984 datum. Updated coordinates that have changed with respect to the shift in datum are provided in the annex.

5 A Study, Port Access Route Study: Approaches to Cape Fear, is referenced in document NCSR 13.8. The study aimed to characterize whether amendments to existing, or additional routeing measures are necessary to improve navigation safety due to factors such as planned or potential offshore energy development, current port capabilities and planned

improvements, increased vessel traffic, existing and potential anchorage areas, changing vessel traffic patterns, meteorological and oceanographic effects, or navigational difficulty in the proposal area. An in-depth analysis of the history of groundings, collisions, or damage to the marine environment can be found in section E of the study. The study recommends establishing a precautionary area at the offshore terminus of the TSS and concludes that its establishment would improve navigation safety in the area.

**Description of the area**

6 The location of the proposed precautionary area (light magenta), existing TSS, United States fairways (light green), and offshore energy areas (blue) are provided in figure 2. Geographic positions, number, edition and geodetic datum of the reference chart used to define the amended routeing system, are set out in the annex.

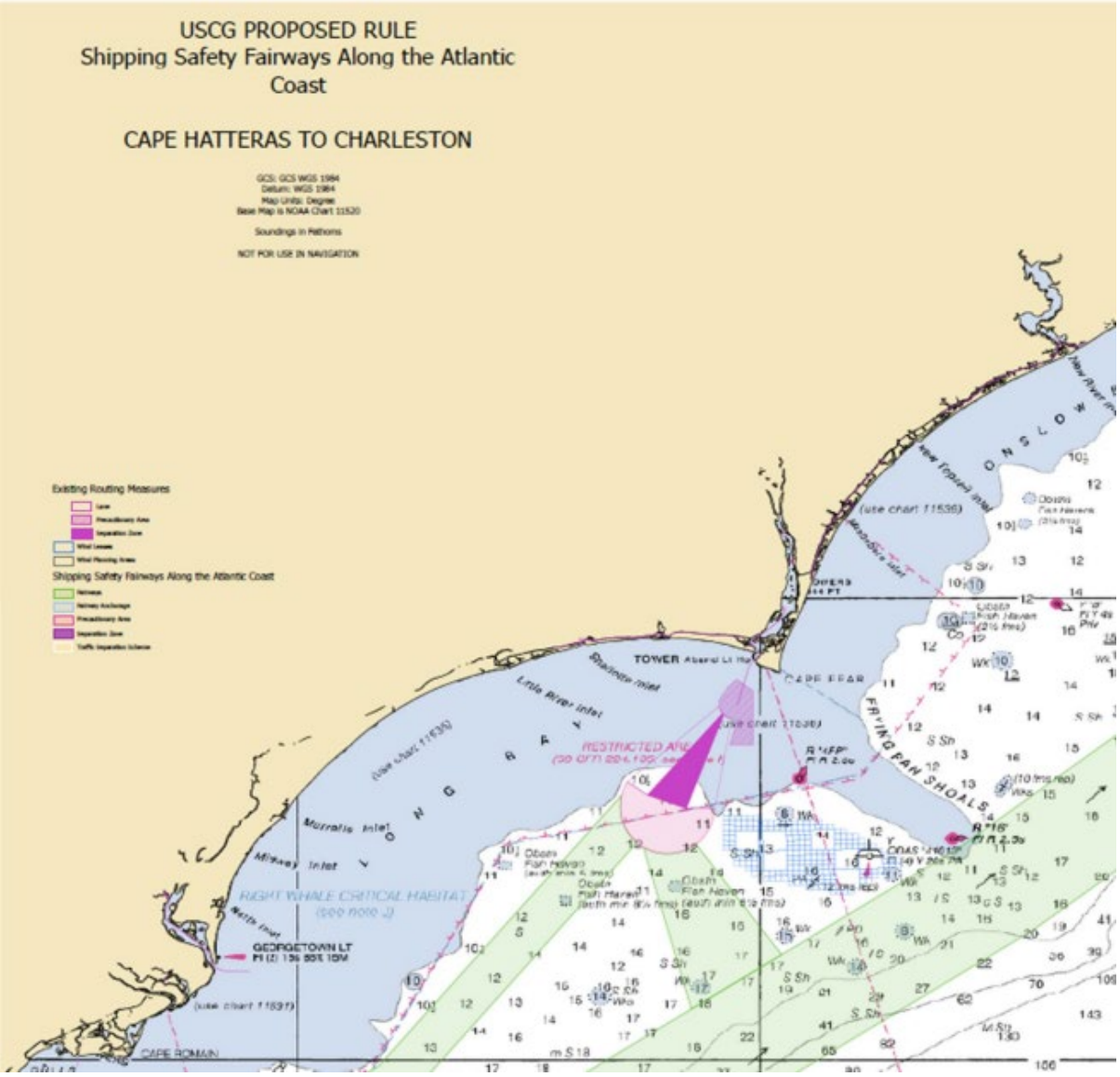



**Figure 2: Proposed precautionary area with existing TSS and U.S. fairways**

**Cooperation between interested States**

7 The United States has all rights, jurisdiction and obligations to the waters in the proposal to amend the TSS in the approaches to the Cape Fear River. The TSS is entirely in the EEZ of the United States, and there is no need for a cooperation agreement with other States.

**Traffic considerations**

8 The United States analysed vessel traffic in the approaches to the Cape Fear River and focused on waters located seaward of existing port approaches within the United States EEZ. The study referenced in document NCSR 13/INF.8 allowed the United States to consider vessel movements among both domestic and international ports of call to inform marine planning in the approaches to the Cape Fear River. Information used included automatic identification system (AIS) data and information from shipping organizations to identify traditional navigation routes. The AIS data identified customary routes transited by vessels from March 2023 through March 2024. Figure 3 provides a graphic of historic vessel tracks in the proposed area.

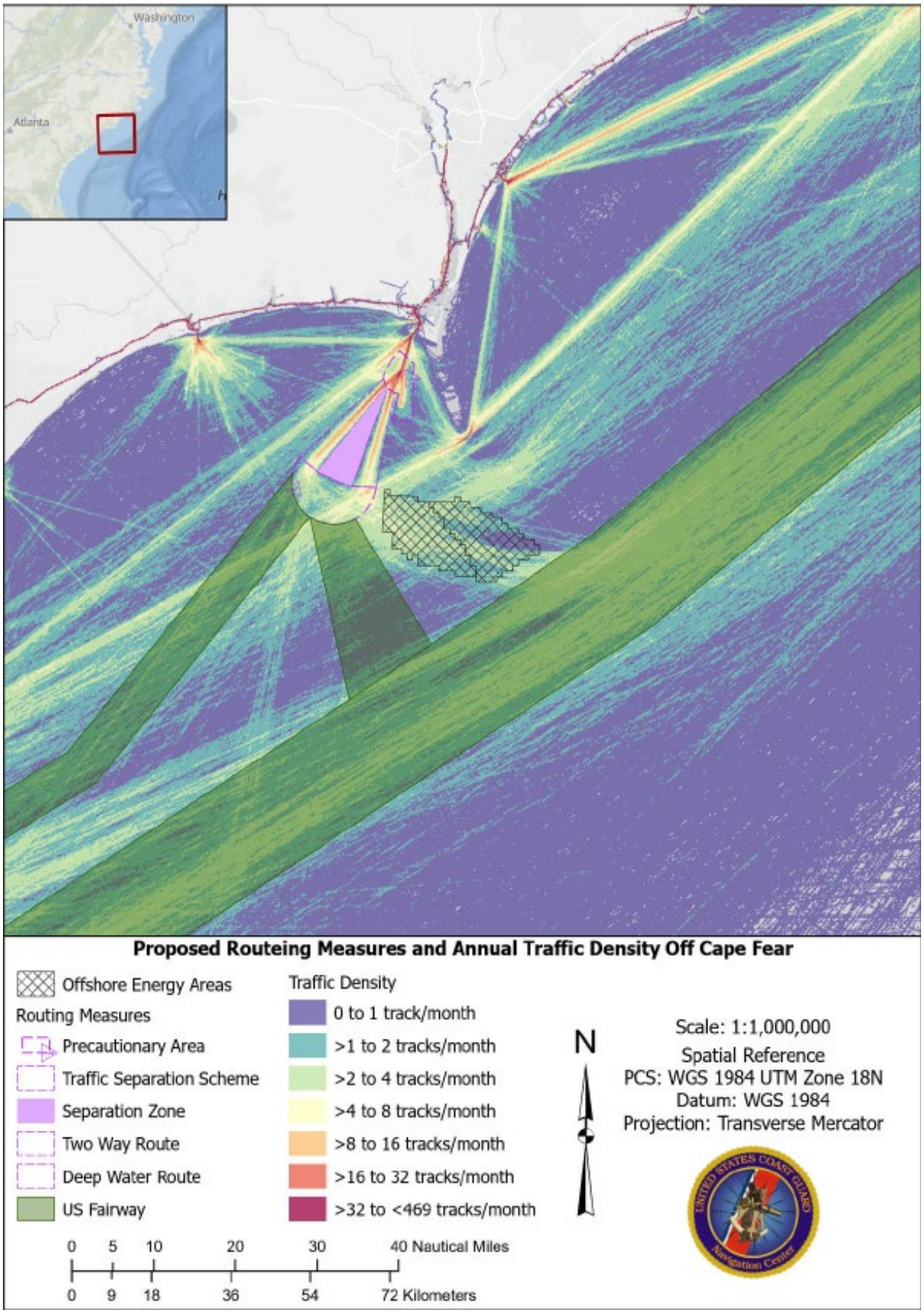



**Figure 3: AIS vessel movements in the approaches to the Cape Fear River**

9 Figure 3 shows the convergence of vessels entering or exiting the TSS from multiple directions. Traffic is expected to divert to the south or north of the offshore energy area. Of concern are vessels transiting past the TSS that typically transit through the new offshore energy area. These vessels are expected to transit to the north of the offshore energy area and converge with the traffic in the proposed precautionary area, increasing the existing traffic density in a confined area. Although only one location has been identified for offshore energy and depicted in figure 3, any area that is not protected by the United States fairway designation or an adopted IMO routeing measure is available for potential development, thus exacerbating the concentration of traffic in the proposed precautionary area.

10 The existing aids to navigation (AtoN) can be seen in figure 4. There are no AtoN in the established TSS. Frying Pan Shoals, to the east of the TSS, is marked with four lateral buoys and one safe water mark "SA", none of which will be impacted by the establishment of the new precautionary area. The "Cape Fear River Entrance Channel" begins with the safe water mark, "CF" which coincides with geographical position 16 (see annex for coordinates) in the approximate centre of the currently established precautionary area A Cape Fear Lighted Research Buoy "A" is not used for navigational purposes and is not impacted by this proposal.

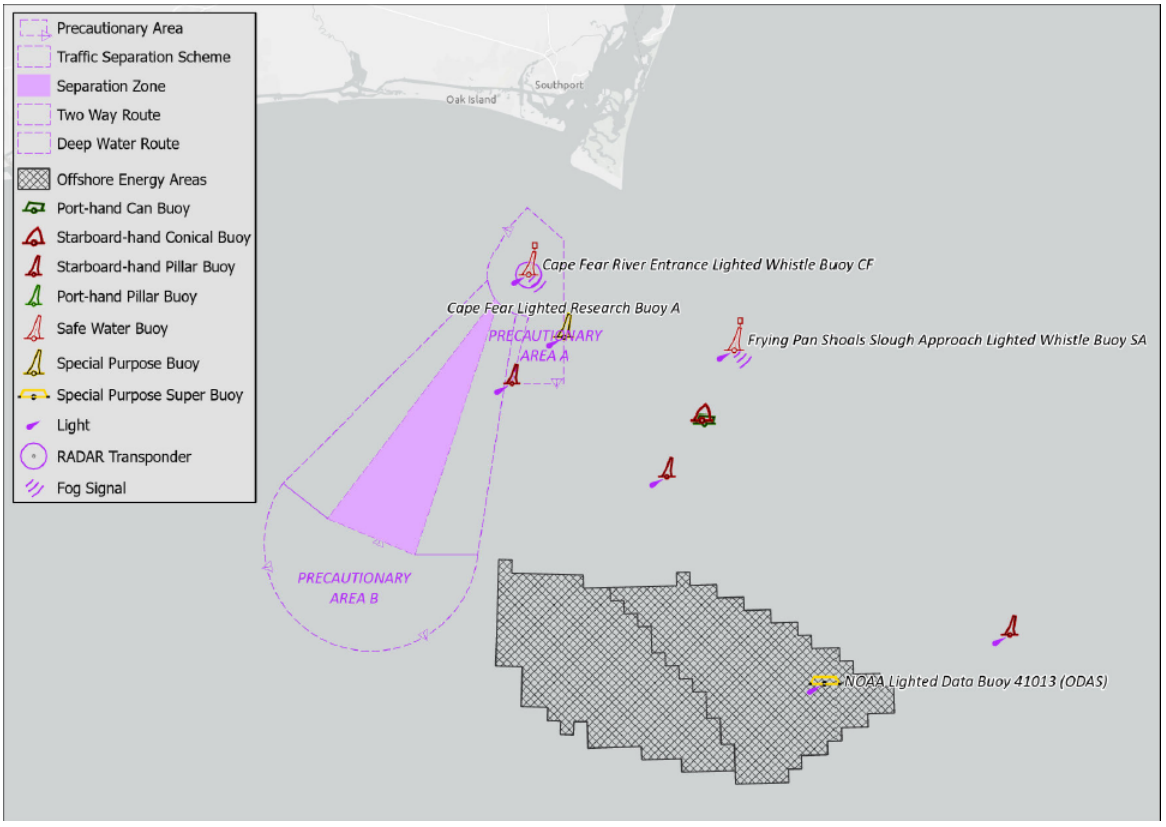



**Figure 4: Existing aids to navigation in the approaches to the Cape Fear River**

11 This proposal is seeking to establish a new precautionary area seaward of the existing TSS. A virtual AtoN safe water mark in the centre of the proposed precautionary area was considered, but the current AIS AtoN infrastructure does not support the ability to broadcast the virtual AtoN in the proper location at this time. Further consideration of AIS AtoN will be considered in the future as the capability improves. Consequently, AtoN in the area do not need to be amended or supplemented.

12 There is no vessel traffic service in the area.

13 The United States National Oceanic and Atmospheric Administration (NOAA) hydrographic surveys for US4NC11M and US3SC10M adhere to International Hydrographic Organization (IHO) standards, including special publication 44. These surveys accurately represent the water depths in the area, and the depths provided on the nautical charts were updated in January 2025. All papers charts in the United States were discontinued by January 2025. Corresponding ENCs are provided for reference and updated in the annex as appropriate.

14 An offshore energy area is located approximately 5 nautical miles (NM thereafter) to the east of the TSS and can be seen in figures 1 and 2. The United States law prohibits the construction of structures within an established TSS.

**Marine environmental considerations**

15 The prevailing wind directions in the area, are along the coast during the summer months and across the coast during the winter months. Average wind speed is 14-18 knots with seas of 2-3 feet. There are diurnal tides with high and low waters occurring at about the same time along the North Carolina seacoast. The currents have considerable velocity in the inlets along the seacoast and in the narrow channels connecting the inlets and the bays and sounds. Historical data from offshore weather buoys indicate a predominant current to flow in an easterly direction. Seasonal severe weather occurs in the late summer and mid-winter with high winds of 28 knots or more from the northwest through north from December through March. Tropical cyclones are a threat throughout the summer and fall with the peak of the season in August and September.

16 The United States considered vessel noise, vessel operations, and vessel movements in the area of the proposal. These considerations are not expected to result in significant impacts to the following resources: air quality, ambient sound, marine vegetation, marine invertebrates, flying insects, birds, bats, marine fish, essential fish habitat, commercial fishing, marine construction, mineral extraction, oil and gas extraction, recreation and tourism, existing energy projects, research, transportation and shipping, and subsistence fishing and hunting.

17 There are no marine environmental considerations that will impact this proposal.

18 Additional information regarding marine and environmental considerations can be found in document NCSR 13/INF.8.

**Status of the routeing system**

19 This proposal is recommendatory and is designed to preserve traditional maritime routes and safe access to the United States ports and protect them from fixed structures that could impact navigation safety and generally prioritize the right of navigation over all other uses.

**Position-fixing in relation to the routeing system**

20 Accurate position-fixing is widely available to all vessel traffic using the global positioning system (GPS). Aids to navigation, including the establishment of virtual AtoN, are addressed in paragraphs 10 and 11. Because of low user numbers and significant operational costs compared to satellite-based GPS, LORAN-C was considered unnecessary for modern navigation and discontinued in February 2010. Nautical charts for the area are available. GPS and nautical charts used in conjunction with AIS provide real-time location data to other vessels and shore stations.

**Miscellaneous information**

21 Prior to establishing the fairways, as discussed in paragraph 3, the United States conducted a Port Access Route study (PARS) to determine traffic density; if current vessel routeing measures are adequate or require modifications; the type of new vessel routeing measures as necessary; and if the usage of the vessel routeing measures should be mandatory for specific classes of vessels. As provided in paragraph 5, a copy of this study is provided in document NCSR 13/INF.8.

22 Mariner participation in a PARS is essential for the United States to gather firsthand operational data, ensuring that proposed, updated, or new fairways and routeing measures safely accommodate vessel traffic while mitigating risks from offshore development and congestion. Cape Fear River pilots, local port authorities, commercial fishing, passenger vessel operators, bulk and container vessel operators, and recreational boaters were consulted throughout the study to capture direct input regarding the proposed area. Pertinent information has been provided in this proposal. Additional information captured from the consultations and subsequent analysis is provided in document NCSR 13/INF.8.

**Proposed date of implementation**

23 It is proposed that the amendment enter into force six months after adoption by the Committee.

**Action requested of the Sub-Committee**

24 The Sub-Committee is invited to consider the proposal in the annex and recommend it to the Committee for adoption.

***

# ANNEX

## PROPOSED AMENDED TRAFFIC SEPARATION SCHEME AND ASSOCIATED MEASURES "IN THE APPROACHES TO THE CAPE FEAR RIVER"

The highlighted areas are the proposed changes.

**IN THE APPROACHES TO THE CAPE FEAR RIVER**

(Reference charts: United States ~~11536, 2003 edition; 11537, 2004 edition~~US4NC11M, 2004 edition; US3SC10M, 2025 edition.

**Note:** These charts are based on ~~North American 1983~~World Geodetic System of 1984 datum.)

**Description of the traffic separation scheme**

(a) A traffic separation zone is bounded by a line connecting the following geographical positions:

| | | | |
|---|---|---|---|
| (1) 33°44′.94 N, | 078°04′.81 W | (3) 33°34′.50 N, | 078°14′.70 W |
| (2) 33°32′.75 N, | 078°09′.66 W | (4) 33°45′.11 N, | 078°04′.~~98~~97 W |

(b) A traffic lane for northbound traffic is established between the separation zone and a line connecting the following geographic positions:

| | | | |
|---|---|---|---|
| (5) 33°32′.75 N, | 078°05′.99 W | (6) 33°44′.38 N, | 078°03′.77 W |

(c) A traffic lane for southbound traffic is established between the separation zone and a line connecting the following geographic positions:

| | | | |
|---|---|---|---|
| (7) 33°36′.22 N, | 078°~~18′.00~~17′.30 W | (8) 33°46′.03 N, | 078°05.41′ W |

**Precautionary area**

(a) ~~A p~~Precautionary area A is established bounded by a line connecting the following geographical positions: from

(9) 33°47′.65 N, 078°04′.78 W; to
(10) 33°48′.50 N, 078°04′.27 W; to
(11) 33°49′.53 N, 078°03′.10 W; to
(12) 33°48′.00 N, 078°01′.00 W; to
(13) 33°41′.00 N, 078°01′.00 W; to
(14) 33°41′.00 N, 078°04′.00 W; to
(15) 33°44′.28 N, 078°03′.02 W

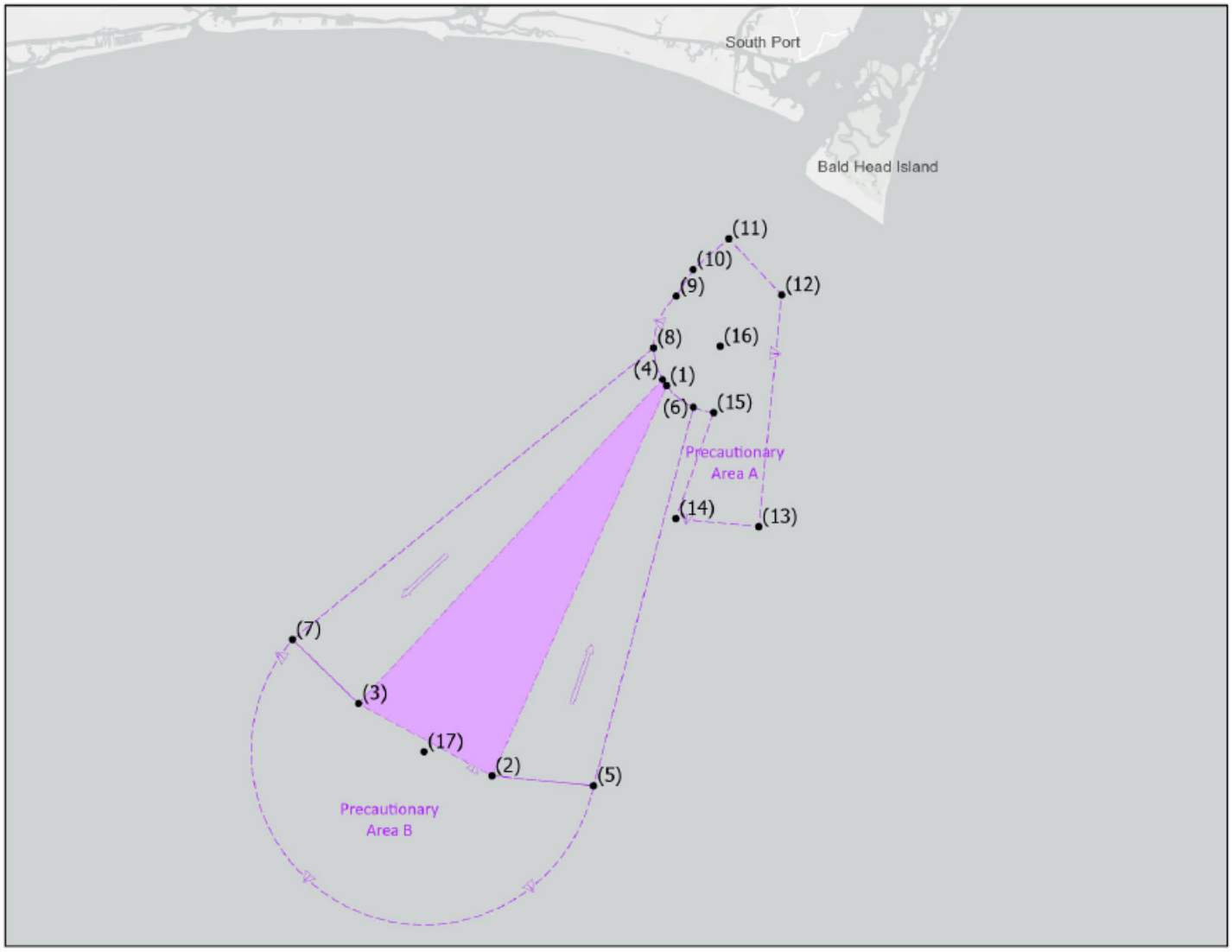
thence by an arc of 2 nautical miles radius, centered at (16) 33°46′.28 N, 078°03′.03 W; thence to the point of origin at (9).

(b) Precautionary area B is established bounded by a line connecting the following geographical positions from:

(7) 33°36′.22 N, 078°17'.30 W; thence easterly by an arc of 5.2 NM centred at (17) 33°32′.26 N, 078°12′.18 W; to
(5) 33°32′.75 N, 078°05′.99 W; thence westerly to
(2) 33°32′.75 N, 078°09′.66 W; thence north-westerly to
(3) 33°34′.50 N, 078°14′.70 W; thence north-westerly to
(7) 33°36′.22 N, 078°17'.30 W.

**Note:** A pilot boarding area is located inside ~~the~~ precautionary area A. Owing to heavy ship traffic, mariners are advised not to anchor or linger in the precautionary area except to pick up or disembark a pilot.

.


South Port
Bald Head Island
(11)
(10)
(9)
(12)
(8)
(16)
(4)
(1)
(6)
(15)
Precautionary
Area A
(14)
(13)
(7)
(3)
(17)
(2)
(5)
Precautionary
Area B